

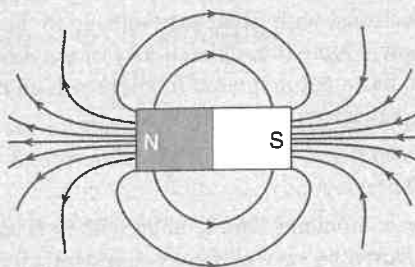
CHAPTER 18 SUMMARY

Magnets, Poles, and Dipoles

(Section 18.1) Magnets have north and south poles; like poles repel, unlike poles attract.

The simplest magnetic field is a **dipole field**; there are no **magnetic monopoles**.

The discrepancy between Earth's magnetic north and geographic north is called **declination**. The angle between the magnetic field vector and Earth's surface is called the **inclination**.



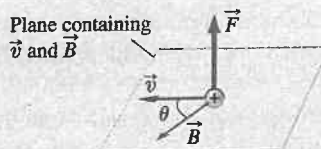
Magnetic Force on a Moving Charge

(Section 18.2) A charged particle in a magnetic field experiences a force perpendicular to the magnetic field and to the particle's velocity.

A charged particle moving perpendicular to a uniform magnetic field follows a circular trajectory.

The direction of the charged particle's motion around the circle depends on the orientation of the magnetic field and the sign of the charge.

Magnetic force: $F = |q|vB \sin \theta$



Radius of circle of charged particle in circular motion:

$$R = \frac{mv}{|q|B}$$

Applications of Magnetic Forces

(Section 18.3) In a **velocity selector**, for any choice of E and B , there is only one velocity of charged particle that passes through without deflection.

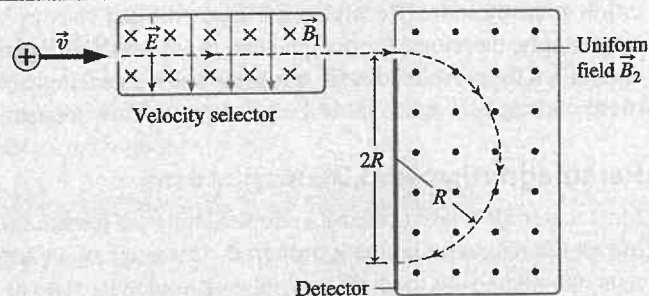
The **mass spectrometer** allows for measurement of a particle's **charge-to-mass ratio** and is good for separating isotopes.

The **Hall effect** can be used to measure magnetic fields or charge carrier density.

Cyclotrons and **synchrotrons** are used to accelerate subatomic particles to high energies.

Charge-to-mass ratio in a mass spectrometer: $\frac{|q|}{m} = \frac{E}{B_1 B_2 R}$

Hall potential difference: $\Delta V_H = \frac{IB}{neb}$



Cyclotron frequency: $f_c = \frac{|q|B}{2\pi m}$

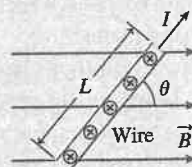
Magnetic Forces on Conducting Wires

(Section 18.4) A current-carrying wire in a magnetic field experiences a force perpendicular to the direction of current and to the magnetic field.

A current loop in a magnetic field experiences torque and has a **magnetic moment**, with a direction given by the right-hand rule.

One application using atomic magnetic moments is **nuclear magnetic resonance (NMR)**, also used in the medical diagnostic technique called **magnetic resonance imaging (MRI)**.

Magnetic force on a conducting wire: $F = ILB \sin \theta$



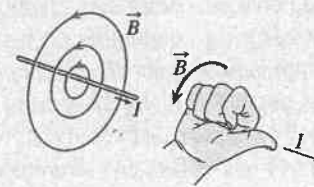
Torque on a current loop: $\tau = NIAB \sin \theta = \mu B \sin \theta$

The Origin of Magnetism

(Section 18.5) Moving charge is the source of a magnetic field, so an electric current produces a magnetic field. The field of a straight wire encircles the wire. A current loop produces a dipole field.

A long cylindrical coil is a **solenoid**, which produces a nearly uniform magnetic field in its interior.

Magnetic field of a long straight wire: $B = \frac{\mu_0 I}{2\pi d}$



Magnetic field at the center of a circular coil: $B = \frac{\mu_0 NI}{2r}$

Magnetic field inside a solenoid: $B = \mu_0 nI$

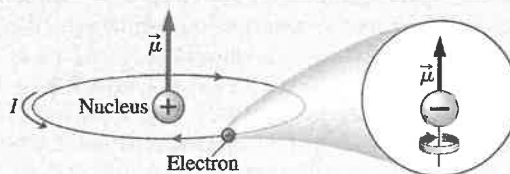
Magnetic Materials

(Section 18.6) Atoms have magnetic properties due to their electrons' orbital motion and spins.

In **ferromagnetic** materials like iron, many atomic moments align to form **magnetic domains**; such materials can have strong magnetic moments.

Paramagnetic materials have a net magnetic moment only in the presence of an applied magnetic field.

Diamagnetic materials respond only weakly to an applied magnetic field, with their magnetic moment opposite the field direction.



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled BIO are of biological or medical interest.

Conceptual Questions

- Explain how and why you expect the magnetic declination to change as you travel eastward from California to New York.
- Explain how and why you expect magnetic inclination to change as you travel southward from Canada to Mexico.
- Suppose that a compass needle in the magnetic field of a strong electromagnet points northeast. What's the field direction at that point?
- The electric field is defined to be in the direction of the electric force on a positively charged particle. Why isn't the magnetic field defined to be in the direction of the magnetic force on a positively charged particle?
- A negatively charged particle is moving horizontally in a region with a magnetic field directed vertically upward. Describe the particle's motion.
- Discuss the trajectory of a charged particle in a magnetic field when the particle's velocity is *not* perpendicular to the field. Explain why the motion is a *helix*, such that the particle's velocity component in the direction of the magnetic field is constant, and the motion in the perpendicular direction is circular.
- A charged particle is injected into a uniform magnetic field. If the angle between the particle's velocity and the magnetic field is 45° , describe the particle's trajectory.
- Does the magnetic force ever do work on a moving charged particle?
- If magnetic fields do no work, is it also true that they don't accelerate charged particles?
- An uncharged particle will always pass through a velocity selector with no deflection. Describe how you could use the velocity selector to determine whether you have a beam of uncharged particles.
- Why are synchrotrons rather than cyclotrons used to accelerate charged particles to extremely high energies?
- Explain why all charged particles with a certain speed pass through a velocity selector undeflected, regardless of their charge and mass.
- You're looking down on a positively charged particle moving in a clockwise circle. In what direction is the resulting magnetic moment? What if it's a negatively charged particle moving in the same direction?
- Explain why a charged particle moving initially perpendicular to a uniform electric field follows a parabolic trajectory, while a charged particle moving perpendicular to a magnetic field follows a circular trajectory.
- A wire carries electric current horizontally from west to east. In what direction should a magnetic field point so that the force on the wire is vertically downward?
- You're at a location in which the magnetic declination is 0° and the inclination is 60° . In what direction is the force on a horizontal wire carrying current (a) from south to north and (b) from west to east?
- A current loop in the x - y plane carries a clockwise current. For each of the following magnetic fields, tell whether the loop will rotate, and if so, in what direction. (a) Magnetic field in $+x$ -direction; (b) magnetic field in $+y$ -direction; (c) magnetic field in $+z$ -direction.
- Describe the force of interaction between two nearby long straight wires that are perpendicular and carry the same current.
- What makes an element a good candidate for each of the following: ferromagnetism, paramagnetism, and diamagnetism?
- Considering the electronic configuration of the five ferromagnetic elements, justify why they are ferromagnetic. Why are some other elements in the same columns of the periodic table not ferromagnetic? Do you expect these other elements to be paramagnetic or diamagnetic?
- You have a fixed length L of wire and wish to make a single loop that will experience the maximum torque in a magnetic field. Is it better to make the coil circular or square, or doesn't it matter?
- A 0.75-T magnetic field points west. Find the magnitude and direction of the force on a $+2.5\text{-}\mu\text{C}$ charge moving southward at 5.5 m/s . (a) $1.0 \times 10^{-5}\text{ N}$, straight up; (b) $1.0 \times 10^{-5}\text{ N}$, straight down; (c) $1.0 \times 10^{-2}\text{ N}$, straight up; (d) $1.0 \times 10^{-2}\text{ N}$, straight down.
- A negative particle traveling northward is deflected toward the east by a magnetic force. The direction of the magnetic field is (a) straight up; (b) straight down; (c) south; (d) west.

Multiple-Choice Problems

24. A charged particle is moving eastward. This particle experiences zero force in a magnetic field directed (a) vertically up; (b) vertically down; (c) north; (d) west.
25. A $-50\text{-}\mu\text{C}$ charge moves vertically upward at 75 m/s . Which magnetic field will produce a northward magnetic force of 15 mN ? (a) 2 T , east; (b) 2 T , west; (c) 4 T , east; (d) 4 T , west.
26. A proton moves at 2.0 Mm/s in a bubble chamber, perpendicular to a magnetic field. The radius of its trajectory is 6.0 cm . The magnetic field strength is (a) 0.25 T ; (b) 0.35 T ; (c) 0.50 T ; (d) 0.60 T .
27. A cyclotron with radius 0.35 m and a uniform 0.14-T magnetic field can accelerate alpha particles to a kinetic energy of (a) 750 eV ; (b) 14 keV ; (c) 42 keV ; (d) 116 keV .
28. A 50-cm -long straight wire carrying 2.5 A makes a 30° angle with a 0.30-T magnetic field. The strength of the magnetic force on the wire is (a) zero; (b) 0.19 N ; (c) 0.28 N ; (d) 0.38 N .
29. A straight conducting wire carries 0.25 A in the $+y$ -direction. Find the magnetic field that results in a force per unit length on the wire of 0.20 N/m in the $+z$ -direction. (a) 1.25 T , $-x$ -direction; (b) 1.25 T , $+x$ -direction; (c) 0.80 T , $-x$ -direction; (d) 0.80 T , $+x$ -direction.
30. A circular wire loop with diameter 24 cm carries 2.5 A . Find the torque on the loop in a 1.5-T magnetic field directed perpendicular to the loop. (a) 0 ; (b) $0.08\text{ N}\cdot\text{m}$; (c) $0.12\text{ N}\cdot\text{m}$; (d) $0.17\text{ N}\cdot\text{m}$.
31. A 20-cm -diameter circular wire loop carries 2.3 A . Find the torque on the loop in a 1.4-T magnetic field directed in the plane of the loop. (a) 0 ; (b) $0.08\text{ N}\cdot\text{m}$; (c) $0.12\text{ N}\cdot\text{m}$; (d) $0.17\text{ N}\cdot\text{m}$.
32. A long straight wire carries 4.5 A . At a distance of 1.0 cm from the wire's center, the magnetic field strength is (a) $4.5 \times 10^{-5}\text{ T}$; (b) $7.7 \times 10^{-5}\text{ T}$; (c) $9.0 \times 10^{-5}\text{ T}$; (d) $2.8 \times 10^{-4}\text{ T}$.
33. A circular coil has 50 turns of wire and carries 2.1 A . If the coil's radius is 3.5 cm , the magnetic field at the coil's center is (a) $3.8 \times 10^{-5}\text{ T}$; (b) $7.6 \times 10^{-5}\text{ T}$; (c) $9.4 \times 10^{-4}\text{ T}$; (d) $1.9 \times 10^{-3}\text{ T}$.

Problems

Section 18.2 Magnetic Force on a Moving Charge

34. ■ A 0.45-T magnetic field points due east. (a) Find the magnitude and direction of the force on a $+0.10\text{-}\mu\text{C}$ charge moving southward at 3.5 m/s . (b) Repeat for a $-0.10\text{-}\mu\text{C}$ charge with the same velocity.
35. ■ A 0.950-T magnetic field points due north. (a) Find the magnitude and direction of the force on a $+1.40\text{-}\mu\text{C}$ charge moving westward at 1.50 m/s . (b) Repeat for a $-1.40\text{-}\mu\text{C}$ charge with the same velocity.
36. ■ A uniform magnetic field points vertically upward. In what direction should a positive charge move so that it experiences a magnetic force toward the (a) north and (b) east?
37. ■ A 1.50-T magnetic field points northeast. Find the magnitude and direction of the force on a $+0.50\text{-C}$ charge that is moving 10 m/s toward the (a) north; (b) east; (c) south; (d) west; (e) southwest.
38. ■■ A $-200\text{-}\mu\text{C}$ charge moves vertically downward at 12.3 m/s . Find the magnetic field (magnitude and direction) required to produce a northward magnetic force of 1.40 mN .
39. ■■ Near Peoria, IL the magnetic declination is zero and the inclination is 70° . (a) Find (a) the magnitude and direction of the force on a proton moving horizontally northward at $2.0 \times 10^5\text{ m/s}$ and (b) the proton's acceleration.
40. ■■ A proton in the upper atmosphere moves at $9.0 \times 10^5\text{ m/s}$, perpendicular to Earth's magnetic field, which at that point has magnitude $5.1 \times 10^{-5}\text{ T}$. (a) Find the curvature radius of the proton's trajectory. (b) Draw the resulting curve with the correct spatial orientation.

Section 18.3 Applications of Magnetic Forces

41. ■■ An electron is moving in the $+y$ -direction at $2.0 \times 10^5\text{ m/s}$ through a uniform magnetic field of 3.40 T in the $-x$ -direction. (a) What are the magnitude and direction of the force on the electron? (b) Find the electric field that would give zero net force on the electron.
42. ■■ A velocity selector has its electric field directed vertically downward, with magnitude 150 N/C . (a) Find the magnitude of the horizontal magnetic field required to select charged particles moving at 300 m/s . (b) This velocity selector is being used on protons. Compute explicitly the magnitudes of the electric and magnetic forces and show that they're equal.
43. ■■ A proton moving at 4.50 km/s in the $+y$ -direction enters a region containing a uniform magnetic field of 450 mT in the $+z$ -direction. Find the electric field that must be applied to give zero net force on the proton.
44. ■■ The positron is a particle with the same mass as an electron but charge $+e$. A positron in a bubble chamber moves perpendicular to a 4.0-T magnetic field. If the positron's kinetic energy is 25 eV , what's the radius of its trajectory in the bubble chamber? Sketch the trajectory, assuming the magnetic field is directed vertically upward.
45. ■ A muon is a subatomic particle with a charge $-e$. A muon moves at $1.5 \times 10^6\text{ m/s}$ through a bubble chamber, its velocity perpendicular to a 1.2-T magnetic field. If the radius of the muon's path is 1.46 mm , what is its mass? Compare your answer with the masses of the electron and proton.
46. ■■ A mass spectrometer is used to separate the uranium isotopes ^{235}U and ^{238}U . (a) What is the ratio of the radii of their circular paths? (b) In this spectrometer, singly ionized uranium ions pass first through a 140-m/s velocity selector, and then into a 2.75-T magnetic field. Find the diameters of the arcs traced out by the two isotopes.
47. ■ A cyclotron of 15.0-cm diameter operates at a frequency of 400 kHz . (a) To what speed can this cyclotron accelerate alpha particles? (b) What's the cyclotron's magnetic field?
48. ■ A singly ionized oxygen atom moves at 9200 m/s in a circular path perpendicular to a 0.75-T magnetic field. Find the radius and period of its circular motion.
49. ■■ A proton takes $95.0\text{ }\mu\text{s}$ to complete one orbit around a 14.5-m -diameter cyclotron. What's the cyclotron's magnetic field?
50. ■■ In a mass spectrometer, singly ionized iron atoms are accelerated through a 120-V potential difference and then injected into a region with a perpendicular magnetic field of 0.42 T . Describe the ions' subsequent motion.
51. ■■ You're designing a bubble chamber with a 1.25-T magnetic field. How large should it be to observe the complete circular path of these particles, each having 1.0 keV kinetic energy: (a) electrons; (b) protons; (c) alpha particles?
52. ■■ You wish to make a mass spectrometer with a chamber 50 cm on a side. How large a magnetic field is needed to keep singly ionized helium atoms (^4He) with kinetic energy 250 eV within the chamber?
53. ■■ What's the magnetic field strength in a microwave oven's magnetron? *Hint:* See the discussion of microwave ovens in Section 18.3.
54. ■■ In 1897, the English physicist J.J. Thomson used crossed electric and magnetic fields to measure the electron's charge-to-mass

ratio. (a) Calculate this ratio, in units of C/kg. (b) Find the charge-to-mass ratio of singly ionized hydrogen, which has the largest such ratio of any ionized atom. (c) Comparing your answers to parts (a) and (b), discuss the implications for the mass of the electron.

55. ■■ The conduction electron density in germanium (a semiconductor) is $2.01 \times 10^{24} \text{ m}^{-3}$. The Hall effect is observed in germanium using a 0.150-mm-thick bar in a magnetic field of 1.25 T. Find the current required in the bar in order to produce a Hall potential difference of 1.0 mV.
56. ■■ A Hall effect experiment uses a silver bar 3.50 μm thick. When the bar carries a current of 1.42 A, a perpendicular magnetic field of 0.155 T results in a Hall potential difference of 6.70 μV . (a) Use these data to determine the density of conduction electrons in silver. (b) How many conduction electrons are there per atom of silver? [Note: The density of silver is 10,490 kg/m^3 .]
57. ■■ A Hall effect probe used to measure magnetic fields is made with a copper bar of thickness 2.00 μm . If the current in the bar is 500 mA, find the perpendicular magnetic field present when the Hall potential difference is (a) 1.0 V; (b) 1.0 mV.

Section 18.4 Magnetic Forces on Conducting Wires

58. ■ A 7.70-cm length of straight wire runs horizontally along a north-south line. The wire carries a 3.45-A current flowing northward through a uniform 1.25-T magnetic field directed vertically upward. Find the magnitude and direction of the magnetic force on the wire.
59. ■■ A copper wire has diameter 0.150 mm and length 10.0 cm. It's in a horizontal plane and carries a current of 2.15 A. Find the magnitude and direction of the magnetic field needed to suspend the wire against gravity.
60. ■■ A square wire loop 20 cm on a side lies in the x - y plane, its sides parallel to the x - and y -axes. The loop has 15 turns and carries a current of 300 mA, clockwise around the loop. Find the net force on the loop when there is a uniform magnetic field of strength 0.50 T (a) in the $+z$ -direction; (b) in the $+x$ -direction; (c) along a diagonal of the square, from lower left to upper right.
61. ■■■ Find the net torque on the loop of the preceding problem for each case described.
62. ■■ A straight conducting wire carries a 7.5-A current in the $+y$ -direction. Find the magnetic field (magnitude and direction) that results in a force per unit length on the wire of 0.14 N/m in the $-z$ -direction.
63. ■■■ A closed wire loop is in the shape of a right triangle, with sides 10 cm, 10 cm, and $10\sqrt{2}$ cm. The loop lies in the x - y plane, with the one corner at the origin and the shorter sides along the x - and y -axes. The loop carries 250 mA in the clockwise direction. A uniform magnetic field of 0.75 T points in the $+z$ -direction. Find the force on each of the three sides of the loop, and add them to get the net force.
64. ■■■ Repeat the preceding problem if the magnetic field is in the $+y$ -direction.
65. ■■ You're at a place where Earth's magnetic field has magnitude $5.0 \times 10^{-5} \text{ T}$ with declination 0° and inclination 70° . Find the force on a 1.0-m length of wire carrying 1.5 A horizontally (a) from south to north and (b) from west to east.
66. ■■ A straight wire carries 2.25 A through a uniform magnetic field of 0.725 T. What angle should the wire make with the magnetic field so that the wire experiences a force per unit length of 1.40 N/m?
67. ■ A 150-turn circular wire loop with a 25-cm diameter carries 1.5 A. Find the coil's magnetic dipole moment.
68. ■■ A 12-cm-diameter circular wire loop carries a current of 2.3 A. Find the torque on the loop in a 1.4-T magnetic field directed (a) perpendicular to the loop and (b) in the plane of the loop.
69. ■■ You have a 1.50-m length of conducting wire, and you wish to make a circular coil with magnetic dipole moment $9.70 \times 10^{-3} \text{ A} \cdot \text{m}^2$ when the loop current is 650 mA. Find the radius of the coil and the number of turns.
70. ■■ An electric motor has a 900-turn coil on its armature. The coil is circular with diameter 12 cm and carries 12.5 A. Find the maximum torque on the coil when it rotates in a 1.50-T magnetic field.
71. ■■ An electric motor has a 1500-turn, 15.0-cm-diameter circular coil on its armature. Find the magnetic field needed to produce a maximum torque of 25.0 N $\cdot$ m when the coil current is 12.0 A.

Section 18.5 The Origin of Magnetism

72. ■ A long straight wire carries 2.5 A along the $+x$ -axis. Find the magnitude and direction of the magnetic field at (a) a point 15 cm away on the $+y$ -axis and (b) a point 15 cm away on the $-y$ -axis.
73. ■■ Two closely spaced parallel wires carry currents of 1.25 A and 1.98 A in opposite directions. Find the magnetic field a distance of 5.0 cm from the pair of wires.
74. ■■ Two long parallel wires each carry 2.5 A in the same direction, with their centers 1.5 cm apart. (a) Find the magnetic field halfway between the wires. (b) Find the magnetic field at a point in the same plane as the wires, 1.5 cm from one wire and 3.0 cm from the other. (c) Find the force of interaction between the wires, and tell whether it's attractive or repulsive.
75. ■■ Repeat the preceding problem if the currents flow in opposite directions.
76. ■■■ A long straight wire carries 5.2 A along the x -axis, in the $+x$ -direction. A second wire carries 5.2 A along the y -axis, in the $+y$ -direction. Find the magnetic field (magnitude and direction) (a) at the point (0.10 m, 0.10 m) and (b) at the point $(-0.10 \text{ m}, -0.10 \text{ m})$.
77. ■■ Two parallel wires with centers separated by 8.6 mm each carry 12 A in the same direction. Find the magnitude and direction of the force per unit length on each wire.
78. ■■ Repeat the preceding problem if the currents are in opposite directions.
79. ■■■ Four long parallel wires pass through the corners of a square 2.0 cm on a side. The wires each carry 2.5 A in the same direction. (a) What is the magnetic field at the center of the square? (b) What is the force per unit length on each wire?
80. ■■■ A long straight wire is held fixed in a horizontal position. A second parallel wire is 2.4 mm below the first but is free to fall under its own weight. The second wire is copper (density 8920 kg/m^3), with diameter 1.0 mm. What equal current in both wires will suspend the lower wire against gravity?
81. ■ Two 8.0-cm-diameter circular wire loops lie directly atop one another. Find the magnetic field at the center of the common circle if each loop carries a current of 7.5 A (a) in the same direction and (b) in the opposite direction.
82. ■ A 25-cm-long solenoid consists of 5000 turns of wire wound uniformly. What current is needed to produce an 8.0-mT magnetic field inside the solenoid?
83. ■■■ In Figure P18.83, the current is 2.0 A. Find the net force on (a) the rectangular loop and (b) the straight wire. (c) Repeat parts

(a) and (b) if the direction of the current in the rectangular loop is reversed.

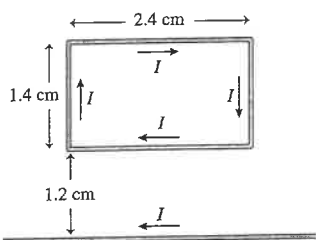


FIGURE P18.83

84. ■ ■ For the situation in Figure P18.84, find the net magnetic field at the center of the circular loop.

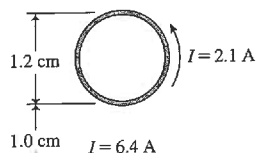


FIGURE P18.84

85. ■ ■ A 20-cm-long solenoid has 4000 turns of wire. Inside this solenoid is a second one, also 20 cm long but with 2500 turns of wire. (a) If a 2.5-A current flows in each solenoid in the same direction, what is the magnetic field inside the inner solenoid? (b) What is the magnetic field inside the inner solenoid if the two carry 2.5-A currents in opposite directions? (c) In each case, what's the magnetic field in the space between the two solenoids?

General Problems

86. ■ ■ A small particle carrying a $+50\text{-}\mu\text{C}$ charge moves at 35 m/s through a uniform 0.75-T magnetic field. The velocity and magnetic field are both in the xy -plane, with the velocity making a 25° angle and the magnetic field a 262° angle, both measured counterclockwise from the $+x$ -axis. Find the magnitude and direction of the magnetic force on the particle.
87. ■ ■ A particle carrying a $-1.4\text{-}\mu\text{C}$ charge moves at 185 m/s through a uniform 45-mT magnetic field pointing in the $+y$ -direction. The resulting force on the particle is $9.3\text{ }\mu\text{N}$ in the $+z$ -direction. In what direction is the particle moving?
88. B10 ■ ■ **Medical cyclotron.** A medical cyclotron with a 2.0-T magnetic field is intended to accelerate deuterium nuclei (deuterons), consisting of one proton and one neutron. (a) At what frequency should the potential difference between the dees be alternated? (b) If the vacuum chamber has diameter 0.90 m, what is the maximum kinetic energy of the deuterons? (c) If the potential difference is 1500 V, how many orbits do the deuterons complete before achieving the maximum energy?
89. ■ ■ A wire of negligible resistance is bent to a rectangle, and a battery and resistor are connected as shown in Figure GP18.89. The right-hand side of the circuit extends into a region containing a uniform 0.15-T magnetic field pointing into the page. Find the magnitude and direction of the net force on the circuit.

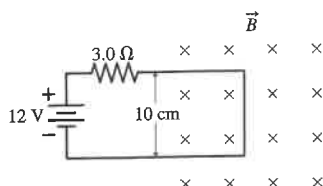


FIGURE GP18.89

90. ■ ■ ■ A rectangular copper bar measures 1.0 mm in the direction of a uniform 2.4-T magnetic field. When the bar carries a 6.8-A current at right angles to the field, the Hall potential difference across it is $1.2\text{ }\mu\text{V}$. Find the number density of free electrons in copper.
91. ■ ■ A single-turn wire loop of radius 3.6 cm carries a 12-A current. It experiences a torque of $0.015\text{ N}\cdot\text{m}$ when the normal to the loop plane makes a 25° angle with a uniform magnetic field. Find the magnetic field strength.
92. ■ ■ A simple electric motor consists of a 100-turn coil 3.0 cm in diameter, mounted between the poles of a magnet that produces a 0.12-T magnetic field. When a 5.0-A current flows in the coil, what are (a) its magnetic dipole moment and (b) the maximum torque developed by the motor?
93. ■ ■ ■ A power line carries a 500-A current toward magnetic north and is suspended 10 m above the ground. The horizontal component of Earth's magnetic field at the power line's latitude is $24\text{ }\mu\text{T}$. If a magnetic compass is placed on the ground directly below the power line, in what direction will it point?
94. ■ ■ You have 10 m of 0.50-mm-diameter copper wire and a power supply capable of passing 15 A through the wire. What magnetic field strengths would you obtain (a) inside a 2.0-cm-diameter solenoid with the wire spaced as closely as possible and (b) at the center of a single circular loop made from the wire?
95. ■ ■ The largest lightning strikes have peak currents around 250 kA, flowing in essentially cylindrical channels of ionized air. How far from such a flash would the resulting magnetic field be equal to Earth's magnetic field strength, about $50\text{ }\mu\text{T}$?
96. ■ ■ You're taking a class in space weather, which involves the far upper atmosphere and magnetic regions surrounding Earth. You're preparing a term paper on the van Allen belts, regions where high-energy particles are trapped within Earth's magnetic field. Your textbook says that the magnetic field strength at the belts is $10\text{ }\mu\text{T}$. To impress your professor, you calculate the radii of the spiral paths of 0.1-MeV, 1.0-MeV, and 10-MeV protons in the van Allen belts. What values do you get?

Answers to Chapter Questions

Answer to Chapter-Opening Question

Auroras result when charged particles from the Sun are trapped in Earth's magnetic field and then enter the atmosphere. Auroras occur because Earth's magnetic field lines emerge predominantly at one pole and return to Earth at the other pole.

Answers to GOT IT? Questions

- Section 18.2 (d)
 Section 18.3 (a) Vertically up
 Section 18.4 (f) Down
 Section 18.5 (c) 4.0 A